ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.6	**Gender**	Male
Height (cm)	175.0	**Weight (kg)**	85.0
BMI	27.8	**Product Info 2**	A1
Occupation	Pension	**Annual Income**	$400,000.00
Insurance Type	Life	**Coverage Amount**	$500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	**Low Risk**
Recommended Annual Premium	**$10,000.00**

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Applicant Details:
* Age: 0.6 years
* Height: 175.0 cm, Weight: 85.0 kg, BMI: 27.8
* Occupation: Pension
* Income: $400,000.0
* Smoker: Yes (1)
* Previous Claims: 0
* Family History: 0
* Insurance Type: Life, Coverage Amount: $500,000.0
* Exercise: Sedentary (0), Alcohol: Moderate/High (3)
* Gender: Male

---

1. Risk Assessment

The applicant presents a highly contradictory risk profile primarily due to an apparent data inconsistency.

Key Risk Factors (assuming *adult* applicant based on other data):

* **Age:** The stated age of **0.6 years** is the most critical and problematic data point. If literal, it indicates an infant. However, the associated `occupation: 'Pension'` and `income: 400,000.0` are logically impossible for an infant. This strongly suggests a data entry error or a misinterpretation of the 'age' variable.
* **BMI (27.8):** Falls into the overweight category, which is a moderate risk factor for various health conditions.
* **Smoking Status (1 - Smoker):** This is a significant adverse risk factor, considerably increasing mortality and morbidity risks for life insurance.
* **Occupation ('Pension'):** While 'Pension' can refer to a retired individual or someone receiving a pension, it is highly inconsistent with an age of 0.6 years. If it implies retirement, the applicant would be an older adult, drastically altering the risk profile.
* **Income ($400,000.0):** High income suggests financial stability, but is irrelevant if the age is 0.6 years.
* **Coverage Amount ($500,000.0):** The coverage amount is reasonable at 1.25 times the stated income, assuming the income is accurate for an adult.
* **Lifestyle (Exercise: 0, Alcohol: 3):** A sedentary lifestyle combined with moderate/high alcohol consumption (depending on scale definition) are negative lifestyle indicators, contributing to increased health risks.
* **Previous Claims (0) & Family History (0):** These are positive factors, indicating no adverse history in these specific areas.
* **Gender (Male):** Generally associated with slightly higher life insurance mortality risks compared to females at certain age bands.

Conclusion: The risk assessment is currently uninterpretable due to the severe age inconsistency. If the applicant is genuinely 0.6 years old, the remaining data (occupation, income, smoking, alcohol) is nonsensical for risk assessment. If 'age: 0.6' is a data error and represents, for example, 60 years, the profile would represent a moderately high-risk adult (smoker, overweight, sedentary, moderate alcohol) with no adverse claims/family history.

2. Comparative Analysis

The Machine Learning Classifier output and RAG context show a strong match to historical cases:

* **Predicted Risk Class:** 1 (lowest risk)
* **Calculated Annual Premium:** $10,000.00
* **Top Similar Cases:** APP-0642B5, POL-59031, POL-46791 all show 100.0% similarity, Risk Class 1, Premium $10,000.00, and 'approved' status.

Analysis of Match/Difference:
The 100.0% similarity reported by the RAG system, alongside the ML model's prediction of Risk Class 1 and the exact premium match, suggests that *all features used for similarity and classification* are identical to these historical approved cases.

This finding creates a critical contradiction with the applicant's `age: 0.6`. If the historical cases are genuinely 100% similar, then they *must have also contained an age of 0.6* (or whatever the underlying representation of `0.6` implies within the system). This raises serious questions about:
1. **Data Integrity:** Whether the 'age' variable is correctly represented or if '0.6' serves as a placeholder/error code that the models are trained on.
2. **Model Robustness:** Why a model trained on such data would classify these profiles as 'Risk Class 1' (lowest risk), especially given the 'smoker' status and other adverse lifestyle factors for an adult, or the complete impossibility of other data for an infant.
3. **RAG System Logic:** How "100% similarity" is determined and if it correctly incorporates all relevant risk factors, particularly age.

Without clarification on the 'age' variable and its interpretation in the historical data, the comparison is misleading.

3. Anomaly & Fraud Detection

Several significant anomalies are present:

* **Primary Anomaly - Age vs. Other Data:** The most glaring anomaly is the `age: 0.6` years. This is utterly inconsistent with `occupation: 'Pension'` and `income: $400,000.0`, which strongly imply an adult applicant, likely retired or nearing retirement. It is impossible for a 0.6-year-old to have an occupation or income, let alone be a smoker or consume alcohol. This suggests either:
* A critical data entry error (e.g., '0.6' intended to be '60').
* A highly unusual internal representation of age, which if true, makes the risk assessment flawed.
* **Inconsistency with ML/RAG Output:** The combination of `smoker: 1`, `bmi: 27.8` (overweight), `exercise: 0` (sedentary), and `alcohol: 3` (moderate/high) for an adult applicant would typically indicate a higher risk profile than 'Risk Class 1'. The classifier's low-risk prediction (and RAG match) is anomalous given these factors, especially if the age is, for example, 60 years. This indicates a potential flaw in the training data or the model's feature importance if it's overlooking these critical risk factors.
* **High Similarity with Anomalous Data:** The 100% similarity to approved historical cases is itself an anomaly, as it implies prior approval for applications with this fundamental data inconsistency or highly unusual profile. This points to potential systemic data quality issues.

No indicators of outright fraud are present based solely on these details, but the severe data inconsistencies warrant a thorough investigation.

4. Premium Recommendation

The calculated annual premium of $10,000.00 is aligned with the ML model's prediction of Risk Class 1 and the historical similar cases. However, this premium recommendation **cannot be reliably justified or accepted** based on the current data.

* **Low Confidence:** The prediction confidence of 55.5% is relatively low for the lowest risk class, which already suggests some uncertainty in the model's assessment.
* **Age Anomaly:** The fundamental error or misrepresentation of the applicant's `age` renders any premium calculation unreliable. If the applicant is, for example, 60 years old (consistent with 'Pension' occupation), a smoker, overweight, and sedentary, a $10,000.00 premium for $500,000.00 life coverage would likely be significantly inadequate for Risk Class 1. Conversely, if it truly is an infant, the other data points are invalid, and the entire premium calculation is baseless.
* **Discrepancy in Risk Factors:** The combination of `smoker: 1`, `bmi: 27.8`, `exercise: 0`, and `alcohol: 3` for an adult applicant strongly contradicts a 'Risk Class 1' assessment. The premium derived from such a classification is likely understated given these adverse factors.

5. Underwriting Decision Recommendation

Refer for Manual Review (Critical)

This application **must be referred for immediate manual review** due to the critical and unresolvable data inconsistency concerning the applicant's age.

Actions Required by Manual Underwriting:

1. **Clarify Applicant Age:** Obtain definitive clarification on the applicant's actual age. This is the paramount step.
2. **Data Validation:** Verify the accuracy of all other associated data points (occupation, income, smoker status, lifestyle) once the age is confirmed.
3. **Re-evaluate Risk Profile:** Based on corrected and validated data, a complete re-evaluation of the applicant's risk factors is necessary.
4. **Re-run Model:** If data is corrected, the machine learning classifier should be re-run to obtain an accurate risk class and premium.
5. **Investigate Similar Cases:** Examine the historical "100% similar" cases to understand how their 'age' variable was handled and if similar data integrity issues exist within the historical dataset.

Proceeding with approval or rejection based on the current data would expose the insurer to significant anti-selection risk, either by insuring a higher-risk individual at an inadequate premium or by incorrectly denying coverage.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.